

ECE NUR BAYRAKTAR

COMPUTER ENGINEERING STUDENT | SOFTWARE & AI ENGINEERING

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PROFESSIONAL SUMMARY

Computer Engineering student building secure full-stack platforms, local RAG systems, machine learning applications, and embedded AI solutions using TypeScript, Python, Java, NestJS, React, PostgreSQL, TensorFlow, and Docker.

EXPERIENCE

Software Engineer Intern | Semafor Teknoloji | August 2026

- Developed CampusOps with NestJS, React, PostgreSQL, Prisma, and Docker; implemented JWT token rotation, RBAC, rate limiting, and audit logging.
- Built state-driven ticket workflows, department assignment, priority/SLA tracking, and ML classification with confidence-based human review.

AI Intern | Microsoft Summer Internship Program | Summer 2026

- Built an offline Phi-3.5 Mini RAG assistant with document chunking, embeddings, hybrid semantic/BM25 retrieval, grounded prompts, and page-level citations.

AI & Software Development Intern | Niso Software Technologies Limited Company | 2026

- Applied data processing, model development, and software engineering practices to production-oriented AI work.

Software Development Intern | Niso Software Technologies Limited Company | 2025 | 25 days

- Developed an ML command recognition application for STM32 deployment under embedded memory constraints.

PROJECTS

CampusOps - AI-Powered Campus Operations Platform | NestJS, TypeScript, PostgreSQL, Prisma, React, Python, ML, JWT, Docker

- Modular REST architecture, Prisma migrations, DTO validation, state-machine workflows, SLA tracking, ML request classification, human review, ADRs, and Dockerized infrastructure.

Local RAG Assistant - Private Document AI | Python, PySide6, Foundry Local, Phi-3.5 Mini, Sentence Transformers, BM25, PyMuPDF

- Privacy-first English/Turkish desktop assistant with local inference, hybrid retrieval, page citations, source inspection, and regression evaluation.

STM32-Based Command Recognition Application | STM32, STM32CubeMX, C, HAL, UART, ADC, CNN, Random Forest, X-CUBE-AI

- Configured GPIO/UART/ADC with STM32CubeMX and HAL; validated the embedded data pipeline through serial output and LED feedback.
- Compared lightweight CNN and Random Forest models, then converted the model with X-CUBE-AI and integrated generated C code for on-device inference.

AI & Machine Learning Projects

- Intel Image Classification: Custom CNN, ResNet50, EfficientNetB0, transfer learning, fine-tuning, Grad-CAM, error analysis.
- House Price Prediction: Feature engineering, cross-validation, LightGBM, XGBoost, CatBoost, hyperparameter tuning, SHAP, residual analysis.
- Titanic & Diabetes: KNN/median imputation, Random Forest, Logistic Regression; from-scratch KNN, normalization, Euclidean distance, Tkinter.

Additional Software Projects

- BookBloom: Spring Boot, PostgreSQL, FastAPI, TF-IDF, cosine similarity, Docker | Dort Bagla Mobile: React Native, Expo, Firebase, offline AI.

EDUCATION

B.Sc. Computer Engineering, 4th Year | Dokuz Eylul University

High School Diploma | Graduated 2nd in School | Hasan Turek Anatolian High School | 2022

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C#, SQL, HTML, CSS

Backend & Applications: NestJS, Node.js, React, Spring Boot, FastAPI, PySide6, React Native, Expo, .NET, REST APIs

AI & Data: TensorFlow, Keras, scikit-learn, RAG, local LLMs, Sentence Transformers, BM25, CNNs, transfer learning, SHAP, NumPy, pandas

Databases & Tools: PostgreSQL, Prisma, MySQL, MSSQL, Docker, Git, JWT, RBAC, testing, database migrations

Embedded AI: STM32, STM32CubeMX, HAL, UART, ADC, X-CUBE-AI

CERTIFICATIONS

Deep Learning Specialization & Generative AI for Software Development (DeepLearning.AI, Aug 2026) | ML Explainability (Kaggle, Aug 2026) | AI Innovators (Microsoft, Aug 2026) | Automotive Camp (OSD, Jul 2026)

LANGUAGES & INTERNATIONAL EXPERIENCE

English: Advanced | German: A2-B1 | Erasmus+ Exchange Program, Poland